



Forward Problem - compute x, y from θ and ϕ

$$x = L_1 \cos \theta + L_2 \cos(\theta + \phi)$$

$$y = L_1 \sin \theta + L_2 \sin(\theta + \phi)$$

Inverse Problem - compute θ and ϕ from x and y

$$x^2 = L_1^2 \cos^2 \theta + L_2^2 \cos^2(\theta + \phi) + 2L_1 L_2 \cos \theta \cos(\theta + \phi)$$

$$y^2 = L_1^2 \sin^2 \theta + L_2^2 \sin^2(\theta + \phi) + 2L_1 L_2 \sin \theta \sin(\theta + \phi)$$

$$x^2 + y^2 = L_1^2 + L_2^2 + 2L_1 L_2 \cos(\phi) \quad \text{because } \cos A \cos B + \sin A \sin B = \cos(A - B)$$

$$\Rightarrow \text{Therefore } \phi = \arccos \left[\frac{(x^2 + y^2) - (L_1^2 + L_2^2)}{2L_1 L_2} \right] \Leftarrow$$

also : $x = L_1 \cos \theta + L_2 \cos \theta \cos \phi - L_2 \sin \theta \sin \phi$

$$x = \cos \theta \{ L_1 + L_2 \cos \phi \} - \sin \theta \{ L_2 \sin \phi \}$$

$$y = L_1 \sin \theta + L_2 \sin \theta \cos \phi + L_2 \cos \theta \sin \phi$$

$$= \cos \theta \{ L_2 \sin \phi \} + \sin \theta \{ L_1 + L_2 \cos \phi \}$$

let $\alpha = \cos \theta$ $k_1 = L_1 + L_2 \cos \phi$

$\beta = \sin \theta$ $k_2 = L_2 \sin \phi$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} k_1 & -k_2 \\ k_2 & k_1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \Rightarrow \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{k_1^2 + k_2^2} \begin{bmatrix} k_1 & k_2 \\ -k_2 & k_1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

so $\alpha = \cos \theta = \frac{1}{k_1^2 + k_2^2} \cdot \{ (L_1 + L_2 \cos \phi) x + (L_2 \sin \phi) y \}$ and $k_1^2 + k_2^2 = x^2 + y^2$

$$\Rightarrow \text{so } \theta = \arccos \left[\frac{(L_1 + L_2 \cos \phi) x + L_2 \sin \phi y}{x^2 + y^2} \right] \Leftarrow$$